

Project Title: University Course Enrollment & Academic Management System

1. Overview

The **University Course Enrollment & Academic Management System** manages student registrations, course catalog administration, enrollment workflows, grade submission, and academic record maintenance. Designed with an MVC architecture, it supports both **Java (Spring MVC)** and **.NET (ASP.NET Core MVC)** implementations.

The system includes five major modules:

1. Student Registration & Profile Management
2. Course Catalog & Semester Offering Management
3. Course Enrollment & Drop/Add Management
4. Faculty Grading & Assessment Management
5. Academic Records & Transcript Generation

2. Assumptions

- Uses **MySQL or SQL Server** for database.
- User roles: **Student, Faculty, Registrar, Admin**.
- ORM frameworks: **Hibernate** (Java) or **Entity Framework Core** (.NET).
- Academic rules (credit limits, prerequisites) are simplified for training.
- No external payment or exam systems integrated.

3. Module-Level Design

3.1 Student Registration & Profile Management Module

Purpose: Handles new student registrations and maintains student profiles.

Controller:

- StudentController
 - registerStudent()
 - updateProfile()
 - getStudentDetails()

Service:

- StudentService
 - Implements student creation, validation, and profile maintenance.

Model:

- Student Entity
 - Attributes:
 - studentId (PK)
 - name
 - email
 - department
 - contactNumber
 - enrollmentYear

3.2 Course Catalog & Semester Offering Management Module

Purpose: Manages the course master list and semester-wise offerings.

Controller:

- CourseController
 - addCourse()
 - updateCourse()
 - getCourseDetails()
 - listCoursesBySemester()

Service:

- CourseService
 - Handles course lifecycle and semester assignment.

Model:

- Course Entity
 - Attributes:
 - courseId (PK)
 - courseName
 - credits

- department
- semesterOffered

3.3 Course Enrollment & Drop/Add Management Module

Purpose: Manages student enrollment requests and add/drop operations.

Controller:

- EnrollmentController
 - enrollCourse()
 - dropCourse()
 - getEnrolledCourses()

Service:

- EnrollmentService
 - Enforces enrollment rules, prerequisites, and seat limits.

Model:

- Enrollment Entity
 - Attributes:
 - enrollmentId (PK)
 - studentId (FK)
 - courseId (FK)
 - enrollmentStatus (ENROLLED, DROPPED)

3.4 Faculty Grading & Assessment Management Module

Purpose: Supports faculty in grading and assessment uploads.

Controller:

- GradeController
 - submitGrade()
 - updateGrade()
 - getGradesByCourse()

Service:

- GradeService
 - Validates grade submissions and stores assessment details.

Model:

- Grade Entity
 - Attributes:
 - gradeId (PK)
 - studentId (FK)
 - courseId (FK)
 - grade
 - remarks

3.5 Academic Records & Transcript Generation Module

Purpose: Maintains academic records and produces transcripts.

Controller:

- TranscriptController
 - generateTranscript()
 - getAcademicRecord()

Service:

- TranscriptService
 - Calculates GPA and compiles academic history.

Model:

- AcademicRecord Entity
 - Attributes:
 - recordId (PK)
 - studentId (FK)
 - courseId (FK)
 - grade
 - semester

4. Database Schema

Student Table

CREATE TABLE Student (

```
studentId INT AUTO_INCREMENT PRIMARY KEY,  
name VARCHAR(100),  
email VARCHAR(100),  
department VARCHAR(100),  
contactNumber VARCHAR(20),  
enrollmentYear INT  
);
```

Course Table

```
CREATE TABLE Course (  
courseId INT AUTO_INCREMENT PRIMARY KEY,  
courseName VARCHAR(100),  
credits INT,  
department VARCHAR(100),  
semesterOffered VARCHAR(20)  
);
```

Enrollment Table

```
CREATE TABLE Enrollment (  
enrollmentId INT AUTO_INCREMENT PRIMARY KEY,  
studentId INT,  
courseId INT,  
enrollmentStatus ENUM('ENROLLED','DROPPED'),  
FOREIGN KEY (studentId) REFERENCES Student(studentId),  
FOREIGN KEY (courseId) REFERENCES Course(courseId)  
);
```

Grade Table

```
CREATE TABLE Grade (  
gradeId INT AUTO_INCREMENT PRIMARY KEY,  
studentId INT,  
courseId INT,  
grade VARCHAR(5),  
remarks VARCHAR(255),  
FOREIGN KEY (studentId) REFERENCES Student(studentId),  
FOREIGN KEY (courseId) REFERENCES Course(courseId)  
);
```

AcademicRecord Table

```
CREATE TABLE AcademicRecord (  
recordId INT AUTO_INCREMENT PRIMARY KEY,  
studentId INT,  
courseId INT,
```

```
grade VARCHAR(5),  
semester VARCHAR(20),  
FOREIGN KEY (studentId) REFERENCES Student(studentId),  
FOREIGN KEY (courseId) REFERENCES Course(courseId)  
);
```

5. Local Deployment Details

- Install MySQL/SQL Server and JDK 17/21/24 or .NET SDK 8.0.
- Deploy using **Tomcat** (Java) or **Kestrel** (.NET).

Steps:

1. Clone repository.
2. Run SQL schema scripts.
3. Configure application.properties or appsettings.json.
4. Build and run application.

6. Conclusion

The University Course Enrollment & Academic Management System offers a modular MVC-based approach to handling student registrations, course offerings, enrollment, grading, and transcript generation. It is ideal for training purposes and academic workflow simulations.